ME551 Vibrations in Linear Systems

Lecture 2

Linear System Theory 1

Chap 1. Linear System Theory

1.1 Problem Formulation of Linear System in Time Domain: $D[x(t)] = f(t)$
1.2 System Classification. Superposition Principle.
1.3 Excitation and Response Characterization
1.4 Response to **Initial Conditions**
1.5 Response to **Harmonic Excitations**: Frequency Response Function
1.6 Response to **Arbitrary Excitations by Laplace Transform**
1.7 Generalized Functions. Response to Generalized Functions
1.8 Response to **Arbitrary Excitations by Convolution Integral**
1.9 State Equations. Response by the Transition Matrix
1.10 Discrete-Time Systems

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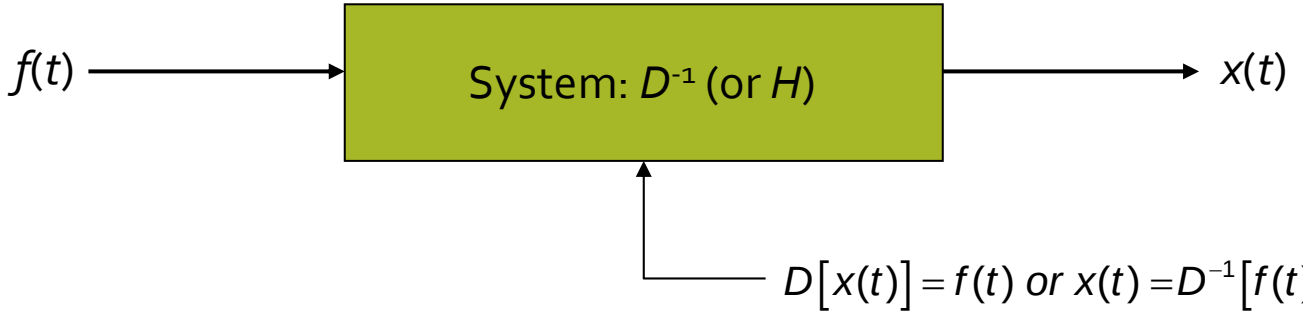
1.1 Problem Formulation of Linear System in Time Domain: D[x(t)] = f(t) (or, x(t) = H[f(t)])

Problem Description:



$D[x(t)] = f(t)$ or $x(t) = D^{-1}[f(t)]$

D: Linear time-invariant differential operator

Problem Definition in ME551= Response Analysis:

Obtain response $x(t)$ when input $f(t)$ and linear differential operator D are given.

(Cf: Input Identification, System Identification are dealt in ME591 Random Data)

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1.2 Definition of System Linearity (D in D[x(t)]); Superposition Principle

D: Linear time-invariant differential operator in this course

- Homegeneity: $D\left[\alpha x(t)\right] = \alpha D\left[x(t)\right]$, where α : constant
- Additivity: $D\left[x_1(t) + x_2(t)\right] = D\left[x_1(t)\right] + D\left[x_2(t)\right]$
- Time-invariance: $D\left[x(t)\right] = f(t) \Leftrightarrow D\left[x(t-\tau)\right] = f(t-\tau)$, *for* $^{A}\tau$

Cf) Superposition principle $\leftrightarrow$ **Linearity = Homogeneity+ Additivity**

$$D\left[\alpha x_1(t) + \beta x_2(t)\right] = \alpha D\left[x_1(t)\right] + \beta D\left[x_2(t)\right]$$

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1.3 Classification of Excitation & Response

$$D\left[x(t)\right] = f(t)$$

$\downarrow \Leftarrow$ Initial conditions

$$x(t) = \underbrace{x_h(t)}_{\text{Due to initial conditions}} + \underbrace{x_p(t)}_{\text{Due to } f(t)}$$

$$x(t)\,\&\,f(t)\begin{cases}\text{Deterministic}\begin{cases}\text{Transient}\\ \text{Harmonic (Sinusoidal)}\\ \text{Periodic}\\ \text{Arbitrary}\end{cases}\\ \text{Random or Stochastic: ME591 Random Data}\end{cases}$$

Notes: 1) Impulsive excitations can be replaced with initial conditions.
2) The system D in most engineering problems is deterministic.

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Initial Condition vs Impulse (a Type of Transient) Excitation (More details in Sec.3.1)

The 1-st order system

$$c\dot{x} + kx = \delta(t), x(-\varepsilon) = 0$$

$$\Updownarrow$$

$$c\dot{x} + kx = 0, x(0) = x_0$$

The 2-nd order system

$$m\ddot{x} + c\dot{x} + kx = \delta(t), x(-\varepsilon) = 0, \dot{x}(-\varepsilon) = 0$$

$$\Updownarrow$$

$$m\ddot{x} + c\dot{x} + kx = 0, x(0) = 0, \dot{x}(0) = \dot{x}_0$$

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1.4 Response to Initial Conditions

$$D[x(t)] \equiv \left(a_0 \frac{d^n}{dt^n} + a_1 \frac{d^{n-1}}{dt^{n-1}} + \ldots + a_n \right) x(t) = 0$$

with initial conditions $x(0), \frac{dx}{dt}(0), \ldots, \frac{d^{n-1}x}{dt^{n-1}}(0)$

$$\downarrow \Leftarrow x(t) = Ae^{st}$$

$$A\left(a_0 s^n + a_1 s^{n-1} + \ldots + a_n\right) e^{st} = 0$$

$$\downarrow \Leftarrow \begin{cases} \textit{Characteristic Equation}: a_0 s^n + a_1 s^{n-1} + \ldots + a_n = a_0 \prod_{i=1}^{n} (s - s_i) = 0 \\ \text{Superposition principle} \end{cases}$$

$$x(t) = \sum_{i=1}^{n} A_i e^{s_i t}, A_i, i = 1, 2, \ldots, n: \ n \text{ unknowns}$$

$$\downarrow \Leftarrow n\ A_i\text{'s from} \begin{cases} n\text{-initial conditions}: \frac{d^k x}{dt^k}(0) = \sum_{i=1}^{n} A_i s_i^k, k = 0, 1, 2, \ldots, n-1 \\ \text{by Gaussian elimination (or LU factorization, Sec.6.1)} \end{cases}$$

$$x(t) = \sum_{i=1}^{n} A_i e^{s_i t}$$

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An Example (n=3):

$$\left(a_0\frac{d^3}{dt^3}+a_1\frac{d^2}{dt^2}+a_2\frac{d}{dt}+a_3\right)x(t)=0 \text{ with initial conditions } x(0),\dot{x}(0),\ddot{x}(0)$$

$$\downarrow\Leftarrow x(t)=Ae^{st}$$

$$A\left(a_0s^3+a_1s^2+a_2s+a_3\right)e^{st}=0$$

$$\downarrow\Leftarrow \begin{cases} a_0s^3+a_1s^2+a_2s+a_3=a_0\left(s-s_1\right)\left(s-s_2\right)\left(s-s_3\right)=0 \\ \text{Superposition principle} \end{cases}$$

$$x(t)=A_1e^{s_1t}+A_2e^{s_2t}+A_3e^{s_3t}, A_i, i=1,2,3: \text{ 3 unknowns}$$

$$\downarrow\Leftarrow A_1,A_2,A_3 \text{ from 3-initial conditions} \begin{cases} x(0)=A_1+A_2+A_3 \\ \dot{x}(0)=A_1s_1+A_2s_2+A_3s_3 \\ \ddot{x}(0)=A_1s_1^2+A_2s_2^2+A_3s_3^2 \end{cases}$$

$$x(t)=A_1e^{s_1t}+A_2e^{s_2t}+A_3e^{s_3t}$$

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1.5 Response to Harmonic Excitations: Frequency Response Function (FRF)

$$\begin{cases} \left(\underbrace{a_0 \frac{d^n}{dt^n} + a_1 \frac{d^{n-1}}{dt^{n-1}} + \ldots + a_n}_{=D} \right) x(t) = F_0 \cos \omega t, \\ \text{where } a_i\text{'s}: \text{real}, \ F_0: \text{real}, \ x: \text{real} \end{cases} \Rightarrow \begin{cases} \mathrm{Re}\left[D\{x(t)\} \right] = \mathrm{Re}\left(F_0 e^{j\omega t} \right), \\ \text{where } x = x + jy: \text{complex} \end{cases}$$

$$\downarrow$$

$$Dx(t) = F_0 e^{j\omega t}, x(t): \text{complex}$$

$$\downarrow \Leftarrow x(t) = X(j\omega) e^{j\omega t}$$

$$X(j\omega) \underbrace{\left(a_0 s^n + a_1 s^{n-1} + \ldots + a_n \right)}_{\equiv Z(s)\text{: Impedance}(\rightarrow \text{dynamic stiffness})} \Bigg|_{s=j\omega} e^{j\omega t} = F_0 e^{j\omega t}$$

$$\downarrow$$

$$G(j\omega) \equiv \frac{1}{Z(s)} \Bigg|_{s=j\omega} = \frac{X(j\omega)}{F_0}: \text{ Compliance or admittance FRF}$$

$$\therefore x(t) = F_0 G(j\omega) e^{j\omega t} \Rightarrow x(t) = \mathrm{Re}\left[F_0 G(j\omega) e^{j\omega t} \right]$$

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Representations of FRF

$$G(j\omega) = |G(j\omega)| e^{-j\phi} : \text{ Magnitude and phase}$$

$$= \mathrm{Re}\left[G(j\omega)\right] + j\,\mathrm{Im}\left[G(j\omega)\right] : \text{ Real and imaginary parts (Cf. polar plot)}$$

Examples for the 1st and 2nd order systems

$$1)\ G(j\omega) = \frac{1}{k + jc\omega} = \frac{1}{\sqrt{k^2 + (c\omega)^2}} e^{-j\tan^{-1}\left(\frac{c\omega}{k}\right)} \Rightarrow |G(j\omega)| = \frac{1}{\sqrt{k^2 + (c\omega)^2}}, \phi(\omega) = \tan^{-1}\left(\frac{c\omega}{k}\right)$$

$$= \frac{k}{k^2 + (c\omega)^2} - j\frac{c\omega}{k^2 + (c\omega)^2} \Rightarrow \mathrm{Re}\left[G(j\omega)\right] = \frac{k}{k^2 + (c\omega)^2}, \mathrm{Im}\left[G(j\omega)\right] = -\frac{c\omega}{k^2 + (c\omega)^2}$$

$$2)\ G(j\omega) = \frac{1}{k - m\omega^2 + jc\omega} = \frac{1}{\sqrt{\left(k - m\omega^2\right)^2 + (c\omega)^2}} e^{-j\tan^{-1}\left(\frac{c\omega}{k - m\omega^2}\right)}$$

$$\Rightarrow |G(j\omega)| = \frac{1}{\sqrt{\left(k - m\omega^2\right)^2 + (c\omega)^2}}, \phi(\omega) = \tan^{-1}\left(\frac{c\omega}{k - m\omega^2}\right)$$

$$= \frac{k - m\omega^2}{\left(k - m\omega^2\right)^2 + (c\omega)^2} - j\frac{c\omega}{\left(k - m\omega^2\right)^2 + (c\omega)^2}$$

$$\Rightarrow \mathrm{Re}\left[G(j\omega)\right] = \frac{k - m\omega^2}{\left(k - m\omega^2\right)^2 + (c\omega)^2}, \mathrm{Im}\left[G(j\omega)\right] = -\frac{c\omega}{\left(k - m\omega^2\right)^2 + (c\omega)^2}$$

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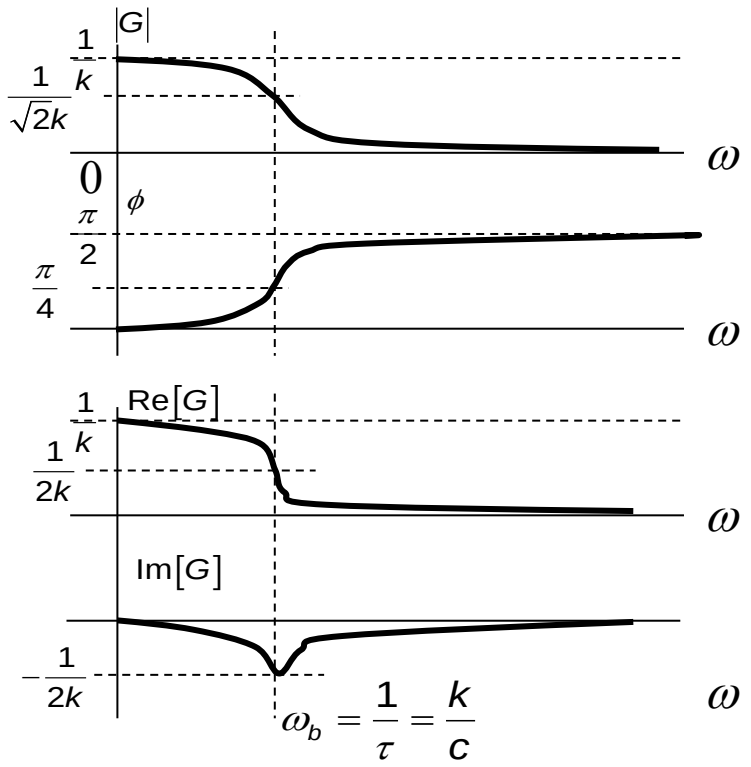
Plots of Compliance Functions

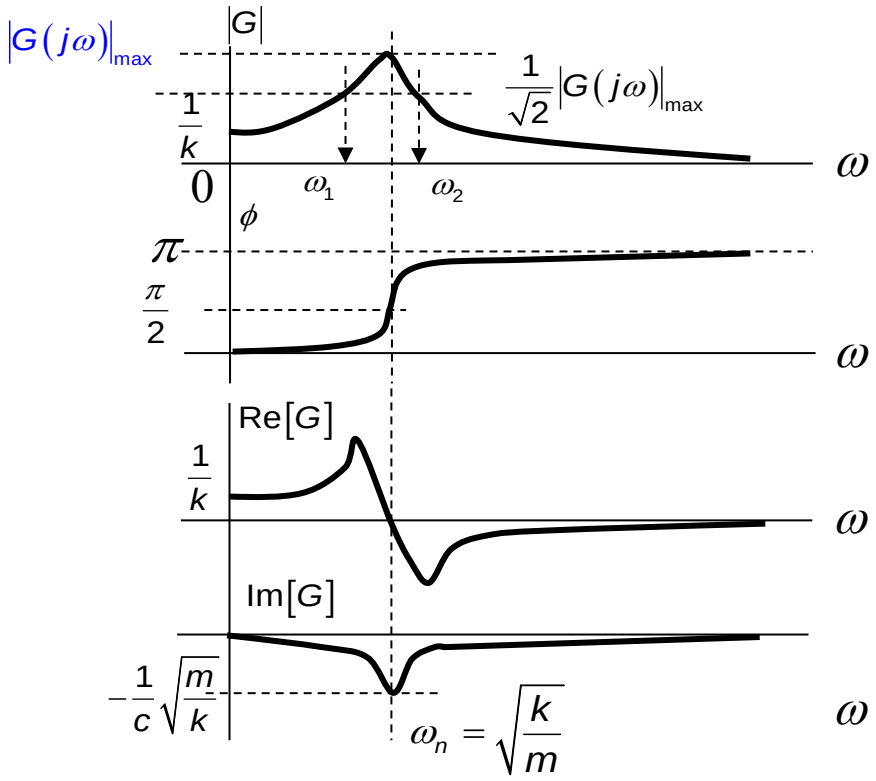
The 1-st order system

$$G(j\omega)=\frac{1}{\sqrt{k^2+(c\omega)^2}}e^{-j\tan^{-1}\left(\frac{c\omega}{k}\right)}=\frac{k}{k^2+(c\omega)^2}-j\frac{c\omega}{k^2+(c\omega)^2}$$



The 2-nd order system

$$G(j\omega)=\frac{1}{\sqrt{\left(k-m\omega^2\right)^2+(c\omega)^2}}e^{-j\tan^{-1}\left(\frac{c\omega}{k-m\omega^2}\right)}=\frac{k-m\omega^2}{\left(k-m\omega^2\right)^2+(c\omega)^2}-j\frac{c\omega}{\left(k-m\omega^2\right)^2+(c\omega)^2}$$



Graphs of the above 2 FRF's can be used for distinction between (or identification of) the 1st and 2nd order systems from measurements!

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SISO System: Single Input Single Output System
MIMO System: Multiple Input Multiple Output System

Various Frequency Response Functions: Definitions by ISO7626-1

Table 1 — Equivalent definitions to be used for various kinds of frequency-response functions related to mechanical mobility

	Motion expressed as velocity	Motion expressed as acceleration	Motion expressed as displacement
Term	Mobility[a]	Accelerance[b]	Dynamic compliance[c]
Symbol	$Y_{ij} = v_i/F_j$	a_i/F_j	x_i/F_j
Unit	m/(N·s)	m/(N·s^2) = kg^{-1}	m/N
Boundary conditions	$F_k = 0;\ k \neq j$	$F_k = 0;\ k \neq j$	$F_k = 0;\ k \neq j$
See	Figure 1	Figure 2	Figure 3
Comment	Boundary conditions are easy to achieve experimentally.		
Term	Blocked impedance	Blocked effective mass	Dynamic stiffness
Symbol	$Z_{ij} = F_i/v_j$	F_i/a_j	F_i/x_j
Unit	N·s/m	N·s^2/m = kg	N/m
Boundary conditions	$v_k = 0;\ k \neq j$	$a_k = 0;\ k \neq j$	$x_k = 0;\ k \neq j$
Comment	Boundary conditions are very difficult or impossible to achieve experimentally (see also A.2).		
Term	Free impedance	Effective mass (free effective mass)	Free dynamic stiffness
Symbol	$F_j/v_i = 1/Y_{ij}$	F_j/a_i	F_j/x_i
Unit	N·s/m	N·s^2/m = kg	N/m
Boundary conditions	$F_k = 0;\ k \neq j$	$F_k = 0;\ k \neq j$	$F_k = 0;\ k \neq j$
Comment	Boundary conditions are easy to achieve, but results shall be used with great caution in system modelling.		

[a] Mobility is sometimes called mechanical admittance.

[b] Accelerance is unfortunately sometimes called inertance. Inertance is not acceptable because it is in conflict with the common definition of acoustic inertance and also contrary to the implication carried by the word inertance.

[c] Dynamic compliance is also called receptance.

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1.6 Response to Arbitrary Excitations by the Laplace Transform

$$D\left[x(t)\right]=f(t),\ \text{where } D=a_0\frac{d^n}{dt^n}+a_1\frac{d^{n-1}}{dt^{n-1}}+\ldots+a_n$$

$$\downarrow\Leftarrow\begin{cases}\text{Laplace transform for zero initial conditions:}\\ X(s)=L\left[x(t)\right]\equiv\int_0^\infty x(t)e^{-st}dt;F(s)=\int_0^\infty f(t)e^{-st}dt\\ L\left[D\left\{x(t)\right\}\right]=\left(a_0s^n+a_1s^{n-1}+\ldots+a_n\right)X(s)\end{cases}$$

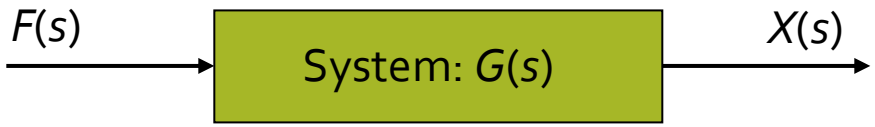
$$\left(a_0s^n+a_1s^{n-1}+\ldots+a_n\right)X(s)=F(s)$$

$$\downarrow\Leftarrow G(s)\equiv\frac{1}{a_0s^n+a_1s^{n-1}+\ldots+a_n}:\ \text{Transfer function of system}$$

$$X(s)=G(s)F(s)$$

$$\downarrow$$

Problem Description:



$$\downarrow$$

$$x(t)=L^{-1}\left[G(s)F(s)\right]=L^{-1}\left[\frac{F(s)}{a_0s^n+a_1s^{n-1}+\ldots+a_n}\right]$$

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$$X(s) = G(s)F(s) = \frac{F(s)}{a_0 s^n + a_1 s^{n-1} + \ldots + a_n} \equiv \frac{N(s)}{D(s)}:$$

$$\mathrm{x}(t) = L^{-1}\left[X(s)\right] \equiv L^{-1}\left[\frac{N(s)}{D(s)}\right],$$ where D for denominator and N for numerator

$\downarrow \Leftarrow$ 1) Solution using Laplace transform table

2) Solution by 1-order factorization of the denominator $D(s)$:

$$D(s) = \prod_{i=1}^{n'}(s - s_i) = (s - s_1)(s - s_2)\ldots(s - s_{n'}), n' \neq n,$$ depending on $F(s)$!

$$= L^{-1}\left[\sum_{i=1}^{n'} \frac{b_i}{s - s_i}\right],$$ where b_i and $s_i, i = 1, 2, \ldots, n'$, are constant.

$$= \sum_{i=1}^{n'} b_i e^{s_i t}$$

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1.7 Generalized (or Singularity) Functions. Response to Generalized Functions

$$\text{Unit impulse (Dirac delta):}\begin{cases}\delta(t-a) = \begin{cases}0 \text{ for } t \neq a \\ \infty \text{ at } t = a\end{cases} \\ \int_{-\infty}^{\infty} \delta(t-a)\,dt = 1\end{cases}$$

$\downarrow \Leftarrow$ Integration

$$\text{Unit step:}\begin{cases}u(t-a) = 0 \text{ for } t < a \\ u(t-a) = ? \text{ at } t = a \\ u(t-a) = 1 \text{ for } t > a\end{cases}$$

$\downarrow \Leftarrow$ Integration

$$\text{Unit ramp:}\begin{cases}r(t-a) = 0 \text{ for } t \leq a \\ r(t-a) = t-a \text{ for } t > a\end{cases}$$

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Responses to 3 generalized functions

$G(s)^{-1} = Z(s)$: impedence

Eq. in time domain | **Solving in *s* domain** | **Response in time domain**

$$D[x(t)] = \delta(t) \underset{\text{Forward}}{\Rightarrow} Z(s)X(s) = 1 \Rightarrow X(s) = G(s) \underset{\text{Inverse}}{\Rightarrow} x(t) = L^{-1}[G(s)] \equiv g(t):\ \text{unit impulse response}$$

$$D[x(t)] = u(t) \underset{\text{Forward}}{\Rightarrow} Z(s)X(s) = \frac{1}{s} \Rightarrow X(s) = \frac{G(s)}{s} \underset{\text{Inverse}}{\Rightarrow} x(t) = L^{-1}\left[\frac{G(s)}{s}\right] \equiv \mathscr{s}(t):\ \text{unit step response}$$

$$D[x(t)] = r(t) \underset{\text{Forward}}{\Rightarrow} Z(s)X(s) = \frac{1}{s^2} \Rightarrow X(s) = \frac{G(s)}{s^2} \underset{\text{Inverse}}{\Rightarrow} x(t) = L^{-1}\left[\frac{G(s)}{s^2}\right] \equiv \mathscr{r}(t):\ \text{unit ramp response}$$

Relation between responses to generalized functions

$$D[g(t)] = \delta(t) \underset{\text{Integration}}{\Rightarrow} \int D[g(\tau)]d\tau = D\left[\int g(\tau)d\tau\right] = u(t)$$

$\downarrow\Leftarrow$ Comparison with step input versus its response: $D[\mathscr{s}(t)] = u(t)$

$$\mathscr{s}(t) = \int g(\tau)d\tau$$

$$D[\mathscr{s}(t)] = u(t) \underset{\text{Integration}}{\Rightarrow} \int D[\mathscr{s}(\tau)]d\tau = D\left[\int \mathscr{s}(\tau)d\tau\right] = r(t)$$

$\downarrow\Leftarrow$ Comparison with ramp input versus its response: $D[\mathscr{r}(t)] = r(t)$

$$\mathscr{r}(t) = \int \mathscr{s}(\tau)d\tau$$

Draw graphs of
for the 1-st and 2-nd order systems!

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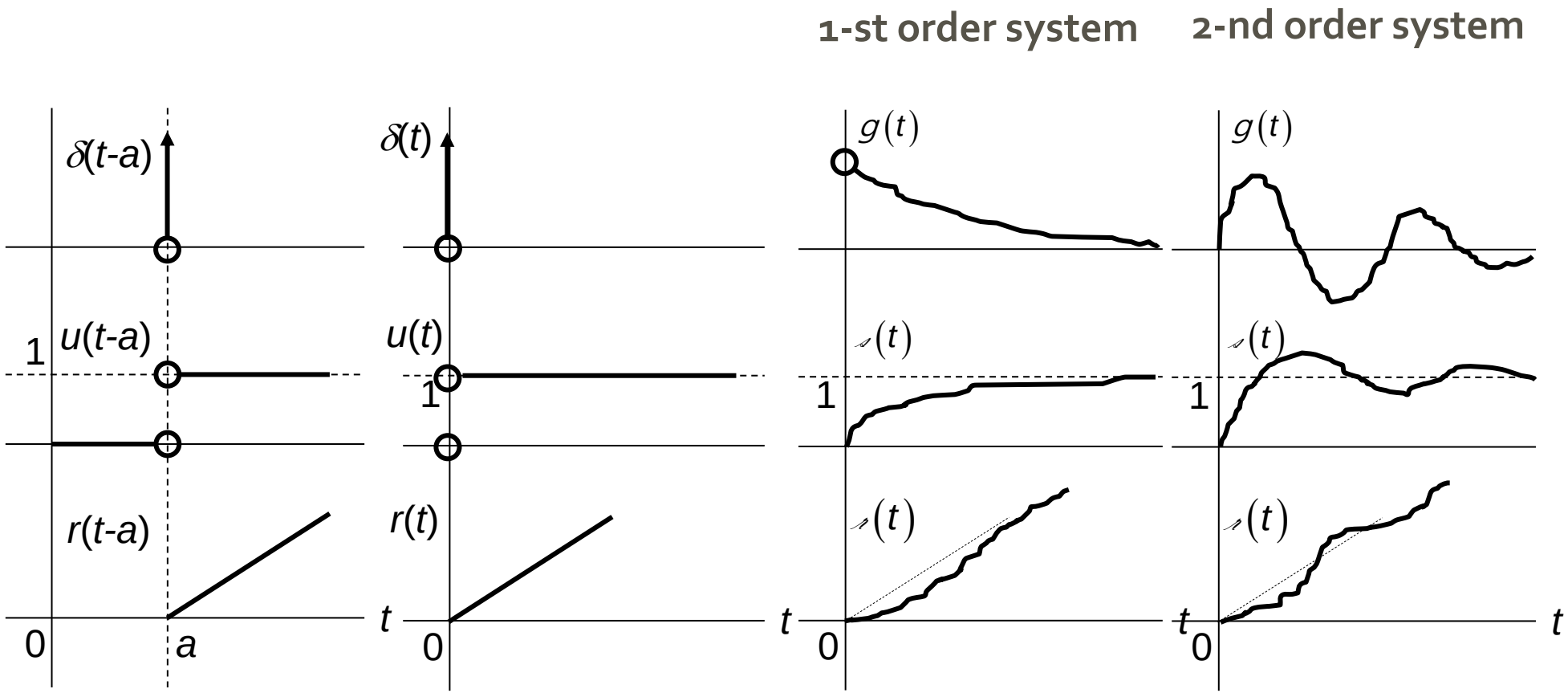
Plots of generalized functions and their system responses



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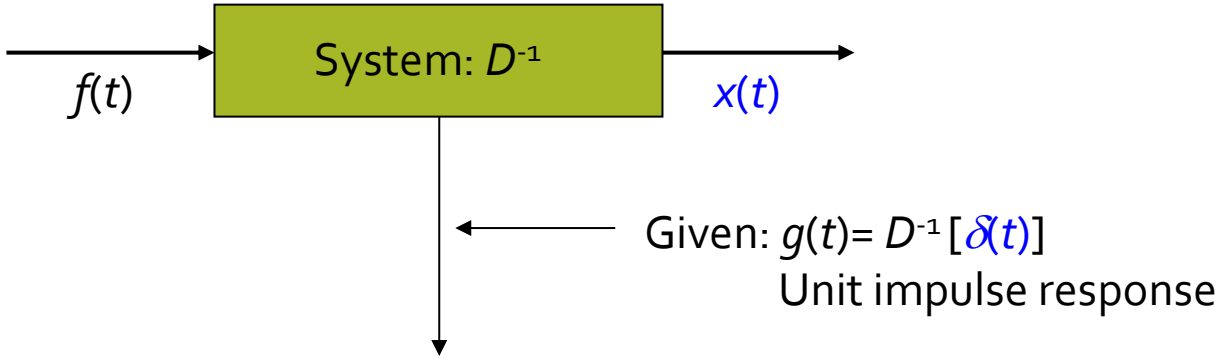
1.8 Response to Arbitrary Excitations by Convolution Integral

Problem Description:



What is the general response $x(t)$ to arbitrary excitation f(t)?

$$x(t) = \int_0^t g(t-\tau)f(\tau)d\tau = \int_0^t g(\tau)f(t-\tau)d\tau :$$ convolution intergal

→ **What is the physical meaning of the convolutional integral?**

$\downarrow \Leftarrow$ Laplace transform; Fourier transform

$$X(s) = G(s)F(s); X(j\omega) = G(j\omega)F(j\omega) \text{ or } X(j\omega) = H(\omega)F(j\omega)$$

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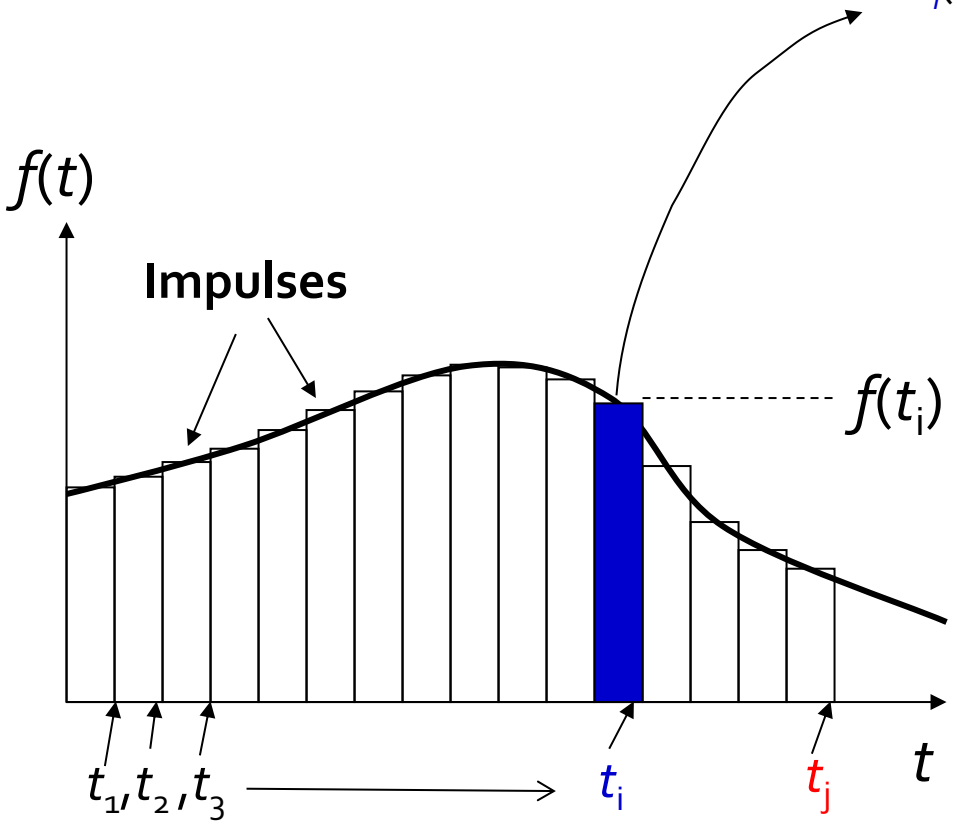
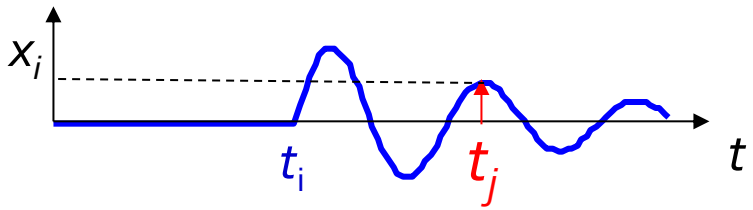
Derivation of convolution integral (physical meaning)

The input force $f(t)$ is viewed as a series of impulses of magnitude $f(t_i)dt$ at time t_i.



A response at t_j to the ith impulse $f(t_i)dt$:
$x_i(t_j) = \{f(t_i)dt\}g(t_j - t_i)$ for $t_j > t_i$



$$x(t_j) = \sum_{i=1}^{j} x_i(t_j) \text{: Total response}$$

$$= \sum_{i=1}^{j} f(t_i)\Delta t \cdot g(t_j - t_i)$$

$$\downarrow \Leftarrow \Delta t \to d\tau, t_i \to \tau, t_j \to t$$

$$x(t) = \underbrace{\int_0^t f(\tau)g(t-\tau)d\tau}_{\text{convolution integral}}$$

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Other forms of convolution integral

$$x(t) = \int_0^t f(\tau)g(t-\tau)d\tau$$

$$\downarrow \Leftarrow t-\tau = \alpha : \tau = t-\alpha; d\tau = -d\alpha$$

$$= \int_t^0 f(t-\alpha)g(\alpha)(-d\alpha)$$

$$= \int_0^t f(t-\alpha)g(\alpha)d\alpha$$

$$\downarrow \Leftarrow \text{The first expression} \Leftarrow f(\tau) = 0 \text{ for } \tau < 0 \text{ and } g(t-\tau) = 0 \text{ for } t < \tau$$

$$= \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau$$

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Laplace transform of convolution integral

$$x(t) = \int_0^\infty f(\tau)g(t-\tau)d\tau : \text{ Convolution integral in time domain}$$

$$\downarrow \Leftarrow \text{Laplace transform}$$

$$X(s) = \int_0^\infty \left\{ \int_0^\infty f(\tau)g(t-\tau)d\tau \right\} e^{-st}dt = \int_0^\infty \int_0^\infty f(\tau)g(t-\tau)e^{-st}dtd\tau$$

$$= \int_0^\infty \int_0^\infty f(\tau)e^{-s\tau}g(t-\tau)e^{-s(t-\tau)}dtd\tau = \int_0^\infty f(\tau)e^{-s\tau} \left\{ \underbrace{\int_0^\infty g(t-\tau)e^{-s(t-\tau)}dt}_{G(s)} \right\} d\tau$$

$$= \int_0^\infty f(\tau)e^{-s\tau}G(s)d\tau$$

$$= G(s)\int_0^\infty f(\tau)e^{-s\tau}d\tau = G(s)F(s) : \text{Multiplication in } s\text{-domain}$$

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Highlight 2

- Linear systems hold principle of superposition (Homogeneity+ Additivity).
- Response to the initial condition, and impulse shows transient behavior related to the system characteristics such as impulse response function.
- Response to harmonic excitation shows steady-state behavior related to the Frequency Response Function (FRF).
- Response to arbitrary excitation can be obtained by either convolution integral of impulse response function and input in the time domain; or multiplication of FRF (TF: Transfer Function) with input in the frequency domain.

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